

Section 15.1: Double Integrals over Rectangles

Multiple Integrals

MTH 310
Calculus III

From Calc I to Calc III

Welcome to **Chapter 15**! We're leaving derivatives behind and entering the world of **multiple integration**.

In Calculus I, the **area** under a curve $y = f(x)$ over $[a, b]$ is the limit of Riemann sums:

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x$$

In Calculus III, the **volume** under a surface $z = f(x, y)$ over a rectangle R is the limit of **double** Riemann sums — using rectangular **columns** instead of rectangular strips.

 Side-by-side comparison. Left panel shows a 2D curve $y = f(x)$ with thin vertical rectangular strips approximating the area underneath. Right panel shows a 3D surface $z = f(x, y)$ with rectangular columns (boxes) of varying heights approximating the volume under the surface over a rectangular region in the xy -plane.

Same idea, one dimension higher: chop the region into pieces, build columns, add up volumes, take the limit.

Building a Double Riemann Sum

Suppose $f(x, y) \geq 0$ is defined on the rectangle $R = [a, b] \times [c, d]$.

Step 1: Partition the rectangle.

- Divide $[a, b]$ into m equal subintervals of width $\Delta x = \frac{b-a}{m}$
- Divide $[c, d]$ into n equal subintervals of width $\Delta y = \frac{d-c}{n}$
- This creates $m \times n$ subrectangles, each with area $\Delta A = \Delta x \cdot \Delta y$

Step 2: Choose sample points.

In each subrectangle R_{ij} , choose a sample point (x_{ij}^*, y_{ij}^*) . Common choices: corners, midpoints, or random points.

Step 3: Build columns and add.

Each column has base area ΔA and height $f(x_{ij}^*, y_{ij}^*)$, so its volume is $f(x_{ij}^*, y_{ij}^*) \Delta A$. The total volume is approximately:

$$V \approx \sum_{i=1}^m \sum_{j=1}^n f(x_{ij}^*, y_{ij}^*) \Delta A$$

 A 3D diagram showing a surface $z = f(x, y)$ over a rectangular region R in the xy -plane. The rectangle is subdivided into a grid of small subrectangles, and over each subrectangle a thin rectangular column rises to the height of the surface at the sample point. The columns together approximate the volume under the surface.

The Double Integral

As we use more and more subrectangles ($m, n \rightarrow \infty$), the approximation gets better and better. In the limit, we get the **exact** volume.

Double Integral over a Rectangle

The **double integral** of f over the rectangle R is:

$$\iint_R f(x, y) dA = \lim_{m, n \rightarrow \infty} \sum_{i=1}^m \sum_{j=1}^n f(x_{ij}^*, y_{ij}^*) \Delta A$$

provided this limit exists (in which case we say f is **integrable** over R).

Good news: If f is continuous on R (or bounded with discontinuities only on a "small" set), then f is integrable over R . All the functions we'll encounter are integrable.

Geometric interpretation:

- When $f(x, y) \geq 0$: $\iint_R f \, dA = \text{volume}$ under the surface and above R
- When f takes both signs: $\iint_R f \, dA = (\text{volume above } R) - (\text{volume below } R)$

The double integral generalizes the single integral: $\int f(x) \, dx$ gives area under a curve; $\iint f(x, y) \, dA$ gives volume under a surface.

Quick Check: Setting Up a Midpoint Riemann Sum

Let $f(x, y) = xy$ and $R = [0, 1] \times [0, 1]$. Partition R into **4 equal squares** ($m = n = 2$).

Write the midpoint Riemann sum approximation to $\iint_R xy \, dA$.

The four subrectangles each have $\Delta A = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$. What are the midpoints?

Example 1: Midpoint Approximation

Example 1

Approximate $\iint_R (x^2 + y^2) \, dA$ where $R = [0, 2] \times [0, 2]$, using $m = n = 2$ (4 equal subrectangles) and **midpoints**.

Each subrectangle is 1×1 , so $\Delta A = 1$. Find the four midpoints and evaluate f at each.

Geometric Interpretation of the Double Integral

When $f(x, y) \geq 0$, the double integral $\iint_R f \, dA$ gives the **volume** of the solid between the surface $z = f(x, y)$ and the rectangle R in the xy -plane.

This means we can evaluate some double integrals by recognizing the solid as a **familiar geometric shape**.

 A rectangular box with base in the xy -plane over the rectangle $R = [1, 7] \times [-2, 2]$ and constant height $z = 5$. The box has length 6, width 4, and height 5.

Constant function: $z = c$ over R is a **box**.

Volume = length $\times$ width $\times$ height

 A wedge-shaped solid (triangular prism) over the rectangle $R = [3, 6] \times [1, 5]$. The surface $z = x$ forms a ramp that rises linearly from $x = 3$ (height 3) to $x = 6$ (height 6). The cross-section in the xz -plane is a trapezoid.

Linear function: $z = x$ over R is a **ramp/wedge**.

Volume = cross-sectional area $\times$ depth

Before reaching for computation, ask: **can I see the shape?** If the solid is a box, wedge, cylinder, cone, or hemisphere, use the geometric formula!

Example 2: Volume of a Box

Example 2

Evaluate $\iint_R 5 \, dA$ where $R = \{(x, y) \mid 1 \leq x \leq 7, -2 \leq y \leq 2\}$.

The surface $z = 5$ is a horizontal plane. What solid sits between this plane and the rectangle R ?

Quick Check: Geometric Interpretation

Without computing any integrals, what is $\iint_R (1 - x) \, dA$ where $R = [0, 1] \times [0, 1]$?

Hint: Sketch $z = 1 - x$ over the unit square. What shape do you see?

The surface $z = 1 - x$ is a plane. At $x = 0$ the height is 1; at $x = 1$ the height is 0. What solid is this?

Example 3: Volume Under a Ramp

Example 3

Evaluate $\iint_R x \, dA$ where $R = \{(x, y) \mid 3 \leq x \leq 6, 1 \leq y \leq 5\}$.

Sketch the surface $z = x$ over R . What is the cross-sectional shape in the xz -plane?

Properties of Double Integrals

Double integrals obey the same **linearity rules** as single integrals.

Theorem: Properties of Double Integrals

If f and g are integrable over R and c is a constant:

1. **Constant multiple:** $\iint_R cf \, dA = c \iint_R f \, dA$
2. **Sum/Difference:** $\iint_R (f \pm g) \, dA = \iint_R f \, dA \pm \iint_R g \, dA$
3. **Comparison:** If $f(x, y) \geq g(x, y)$ for all (x, y) in R , then $\iint_R f \, dA \geq \iint_R g \, dA$
4. **Splitting:** If $R = R_1 \cup R_2$ where R_1 and R_2 don't overlap (except on edges), then $\iint_R f \, dA = \iint_{R_1} f \, dA + \iint_{R_2} f \, dA$

These properties follow directly from the corresponding properties of finite sums and limits. They'll be essential when we learn to **compute** double integrals via iterated integrals in the next section.

Linearity is your best friend: it lets you break complicated integrals into simpler pieces.

Looking Ahead: How Will We Actually Compute These?

So far, we've only evaluated double integrals by:

- Riemann sum approximations (Example 1)
- Geometric formulas for simple shapes (Examples 2 and 3)

The big question: What about $\iint_R e^{x+y} \, dA$ or $\iint_R \sin(x^2y) \, dA$? We can't draw those as boxes or wedges!

Coming in Section 15.2: Fubini's Theorem tells us we can compute a double integral as two nested single integrals:

$$\iint_R f(x, y) \, dA = \int_a^b \left[\int_c^d f(x, y) \, dy \right] dx$$

This is called an **iterated integral**, and it reduces the problem to two Calc I integrals!

Today's takeaway: understand **what** a double integral is (limit of Riemann sums, volume under a surface). Next time: learn **how** to compute them efficiently.

Homework

Section 15.1: 1-11 odd, 17, 19, 21, 25, 29, 31, 33, 45, 55